

Kudrati Khoda

M.Tech Quality, Reliability and Operations Research
Indian Statistical Institute, Kolkata
Mob. +91 8759363639 | kudratikhoda12@gmail.com

Course	College/University	Year	CGPA/%
M.Tech (QR & OR)	Indian Statistical Institute, Kolkata	2025-27	71.3%
B.E. (Electrical Engineering)	Jadavpur University	2025	6.88
Senior Secondary	Hallyan High School (WBCHSE)	2019	81.2%
Secondary	Akhriganj High School (WBBSE)	2017	74.7%

PROJECTS

- Credit Risk Modelling & Portfolio Stress Testing** Python, scikit-learn, LightGBM
LendingClub loan data, 2007–2018
 - Built PD models (Logistic Regression baseline, LightGBM challenger) on **1.34M resolved loans**, using only features available at origination (no leakage); out-of-time (2017) **AUC 0.70 / 0.71**, KS 0.29 / 0.30.
 - Validated across 2015–2018 vintages with decile calibration, aggregate backtesting and PSI (≤ 0.027); identified systematic PD underprediction in 2017 (**19.9% predicted vs 23.1% observed**, ~5.4K excess defaults).
 - Combined PD, realized LGD (93.3%) and EAD to estimate portfolio **Expected Credit Loss of \$494.7M**; simulated correlated defaults (one-factor Gaussian, 10K loans $\times$ 2K scenarios) to obtain **VaR₉₉ \$68.5M** and **ES₉₉ \$74.6M**.
 - Stress and sensitivity testing: PD odds $\times 1.5$ with LGD +5 pp raised VaR₉₉ to \$83.0M; raising default correlation ρ from 0 to 0.5 increased VaR₉₉ **3.6 $\times$** (\$30.5M $\rightarrow$ \$108.4M), showing dependence as the key driver of tail risk.
- Interpretable Survival Risk Prediction using CoxKAN** Python
Guide: Prof. [Supervisor Name]
 - Implemented CoxKAN (Kolmogorov–Arnold Networks) for right-censored time-to-event risk prediction and benchmarked it against Cox Proportional Hazards and DeepSurv.
 - Learned nonlinear hazard functions with interpretable B-spline activations, enabling feature selection and symbolic recovery of the risk equation.
 - Evaluated with the Concordance Index (C-index) and survival prediction metrics.
- Time-to-Event Prediction using KAN-AFT** Python
Guide: Prof. [Supervisor Name]
 - Developed a KAN-based Accelerated Failure Time model and compared it with Weibull, Log-Normal AFT and DeepAFT.
 - Studied nonlinear survival-time modelling and the effect of censoring adjustments (IPCW, Buckley–James).
 - Validated on synthetic, clinical and **financial** datasets using the C-index.

TECHNICAL SKILLS

- Programming:** Python, SQL **Libraries:** Pandas, NumPy, scikit-learn, LightGBM, SciPy **Tools:** Excel, Power BI
- Credit Risk & Validation:** PD / LGD / EAD, Expected Credit Loss, AUC / KS / Gini, Brier score, calibration, backtesting, PSI, VaR, Expected Shortfall, Monte Carlo simulation, stress testing
- Statistics & ML:** Regression, Classification, Hypothesis Testing, Survival Analysis (Kaplan–Meier, Log-Rank, Cox PH, AFT, IPCW), MLP, KAN

RELEVANT COURSES

Survival Analysis | Statistical Quality Control | Reliability Engineering | Stochastic Processes | Multivariate Statistics | Statistical Methods | Probability Theory

POSITIONS OF RESPONSIBILITY

- JU Reunion Convener | Jadavpur University
- Hostel Mess Committee Member

EXTRACURRICULAR ACTIVITIES

Biking | Traveling | Card Games